

and connections created by a circuit editor or circuit design tool. Now, do not confuse IRSIM with a design tool or an editor, as it can simulate but cannot create a netlist. Although it is

an independent tool, it easily accepts a netlist from other tools.

Circuit behaviour and IRSIM

IRSIM being a switch-level simulator works on three important aspects of the behaviour of a digital circuit, namely, transistor state, logic value and transition events.

A transistor state is a linear model state, where

each switch has two states: on and off. This linear model helps IRSIM reflect the differences between the different types of transistors, rise and fall times of the gate and different fabrication processes very accurately.

On the other hand, the logic value behaviour of the circuit allows IRSIM to treat voltages in a circuit as three different values, that is, high (h,1), low (l,0) and indeterminate (unknown or x). This way, IRSIM does not have to bother about the actual voltage that is powering the circuit.

The linear model works with these voltage values and helps compute the generation of a normalised (high) voltage.

The above two aspects help us identify the simplified modelling of a circuit using IRSIM. However, on account of transition events, IRSIM treats a circuit as a cascade of events. This makes IRSIM an event-driven simulator. Let us understand how.

All switch-model transistors have a source-to-drain path and a controlling gate

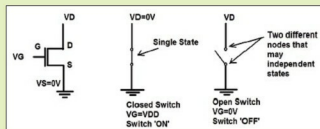


Fig. 3: NMOS transistor acting as a switch

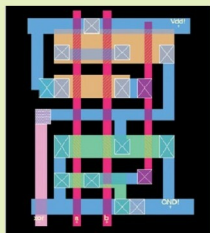


Fig. 4: Layout of 2-input XOR gate

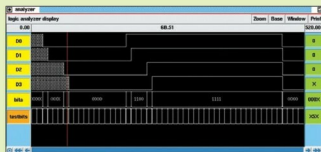


Fig. 5: The analyser graphic display in IRSIM 9.7

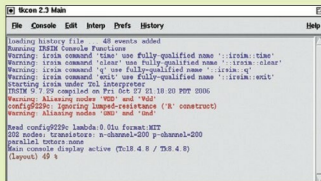


Fig. 6: The IRSIM command console in Tcl or Tk based version 9.7

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